

The Lee–Wick ghost of Yang–Mills mechanics is unstable at every energy:

parameter-free instability laws and an area-filling escape landscape

Joseph Foe
foe.joseph@gmail.com

October 3, 2026

Abstract

Quadratic gravity is renormalisable but propagates a massive spin-2 ghost, and every proposed quantum cure (fakeons, Lee–Wick prescriptions, Merlin modes) inherits a prior classical question: does the interacting Ostrogradsky ghost run away? We study the closest tractable analogue in which the interactions are fixed by gauge symmetry rather than chosen: SU(2) Lee–Wick Yang–Mills theory on a spatial three-torus, restricted to its spatially homogeneous (zero-momentum), colour-diagonal sector, an exact twelve-dimensional Hamiltonian system consisting of chaotic Yang–Mills mechanics coupled to a wrong-sign oscillator. The ghost-free manifold is invariant, and its transverse Lyapunov exponent $\lambda_{\perp}(E)$ is positive at every energy tested, down to $E = 0.008$ in units $g = M = 1$, with no threshold and no exponential suppression. At low energy the growth is parametric resonance driven by the transverse oscillations of the chaotic flow inside the Yang–Mills channels, and we derive the resulting law: the uniform microcanonical measure on the two transverse channel actions, together with adiabatic in-channel motion, gives a u^{-6} spectral tail (measured exponent -6.0) and hence $\lambda \propto E^{9/4}$, with the prefactor fixed by the classical density of states, $\lambda_{\text{diag}} = (32\pi^3/105\sqrt{2}) E^{9/4}/Z = 0.0120 E^{9/4}$ with $Z = 557.1$. The single-mode rate measured directly at $E = 0.008, 0.015, 0.02$ and 0.03 is $(0.0120 \pm 0.0003) E^{2.24 \pm 0.03}$, i.e. the prediction is reproduced at each energy to within 3% with no adjustable parameter, and to within 10% up to $E = 0.1$; within the diagonal sector the physical three-component ghost lies a factor 1.7–2.3 above it over the same range. The same derivation, with one transverse action, predicts $\lambda \propto E^2$ for the two-field x^2y^2 model and is confirmed there to 3–5%. Linearising the full homogeneous theory about the diagonal background shows that the six ghost components the ansatz discards are the faster ones: their exponent exceeds the diagonal law by a factor that reaches 350 at $E = 0.01$, with a shallower energy dependence that is not yet derived, while with the $(D^{\mu}F_{\mu 0})^2$ term of the Lee–Wick action switched off their law is derived, $0.0210 E^{5/4}$. Nonlinearly the ghost escapes to a finite-time singularity through a self-accelerating tachyonic loop, and the escape-time landscape over initial conditions is fractal with uncertainty exponent $\alpha \simeq 0.05$, i.e. boundary dimension $\simeq 1.95$, on every section examined and down to 10^{-14} of the section width: the ghost-free manifold is a repeller with no open basin. We discuss what this implies for the classical limit of the fakeon prescription and for the epoch in which such a ghost could have mattered cosmologically.

1 Introduction

Perturbative quantum gravity built on the Einstein–Hilbert action is not renormalisable; adding the curvature-squared invariants with dimensionless coefficients makes it so [1], at the price of a massive spin-2 field with a wrong-sign kinetic term. Quantised as an ordinary particle this ghost forces a choice between negative norms and an unstable vacuum, and a number of programs attempt to keep Stelle’s renormalisability without the pathology: Anselmi’s fakeons [2, 3], in which the ghost is quantised with an averaged $\pm i\epsilon$ prescription and never appears in asymptotic states;

Lee–Wick-type prescriptions [5, 6]; the Merlin modes of Donoghue and Menezes [7]; agravity [8]. All of these are statements about the quantum theory. They inherit a classical question: any non-degenerate higher-derivative Lagrangian has an Ostrogradsky Hamiltonian unbounded below, and the naive expectation is that interactions let energy flow without limit between the ghost and the normal sector. Recent work has shown that this expectation is not a theorem: interacting ghost systems can be globally stable or “benign” [13, 14, 15], but essentially all such examples use scalar toy models whose interactions were chosen to be stable, and several are integrable or close to it.

Here we study the classical question in a system where the interactions are dictated by a gauge symmetry and the normal sector is chaotic. Lee–Wick Yang–Mills theory,

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu} + \frac{1}{2M^2}(D^\mu F_{\mu\nu})^a (D_\rho F^{\rho\nu})^a, \quad (1)$$

is the gauge-theory counterpart of the Weyl-squared term (the same operator appears in the Lee–Wick Standard Model [6]); it propagates a massive spin-1 ghost of mass M alongside the massless gluon. We place the theory on a spatial three-torus T^3 of side L , so that spatially homogeneous fields are the zero-momentum sector of a finite-volume system, the setting of Lüscher’s small-volume analysis [20], rather than an idealisation in infinite space. Restricted to that sector in the colour-diagonal ansatz the theory reduces to an exact twelve-dimensional Hamiltonian system [17]: Yang–Mills mechanics [18, 19] (the $D = 4$ analogue, with three matrices, of the bosonic sector of the BFSS matrix model at $N = 2$), which is chaotic on almost all of phase space at every energy (the regular islands [21] have small measure, and by the scaling symmetry the same fraction at every E), coupled to a three-component ghost. Because the sector is exact, instability here is an existence proof of instability of the full classical theory on the torus (the converse would not have held).

Our results are as follows. (i) The ghost-free manifold is invariant and, within the colour-diagonal sector, its transverse Lyapunov exponent is positive at every energy tested, down to $E = 0.008$ in the natural units, following a power law close to $E^{9/4}$ at low energy (Sec. 4). (ii) The exponent $9/4$ and the prefactor of the single-mode (“diagonal”) rate are derived from the channel statistics of Yang–Mills mechanics and the classical density of states, with no free parameter, and the derived law $0.0120 E^{9/4}$ is reproduced by direct measurement to within 3% at four energies (Sec. 5). In particular the absence of exponential suppression follows from the microcanonical measure on small transverse actions being a power law. (iii) Nonlinearly the ghost escapes to a finite-time singularity through an identified mechanism, and the escape-time landscape over initial conditions is fractal with boundary dimension $\simeq 1.95$ on every section examined (Sec. 6). (iv) On periodic orbits the ghost is Floquet-stable in most of the energy range, including energies at which its mass matrix is transiently negative (Sec. 7); these are the only exact invariant sets on which a ghost can sit, and they are unstable within the base flow. (v) The derivation generalises to n transverse actions with exponent $(n + 7)/4$ and is confirmed on the two-field model ($n = 1$) (Sec. 5.4); and linearising the full homogeneous theory about the diagonal background shows the off-diagonal Yang–Mills directions to be neutral and the six off-diagonal ghost components to grow faster than the diagonal ones, by a factor 350 at $E = 0.01$, with a law that is derived when the $(D^\mu F_{\mu 0})^2$ term is switched off and measured, but not derived, for the theory itself (Sec. 8). Sec. 9 discusses the implications for the fakeon classical limit and the cosmological scaling of the instability rate.

2 The model

On T^3 with periodic boundary conditions, with $A_i^a = \delta_i^a q_i(t)$ in temporal gauge and $A_0 = 0$, the Gauss-law constraint and the six off-diagonal colour equations vanish identically on the ansatz,

symmetric criticality holds, and the action is $L^3 \int dt L_{\text{red}}$ with [17]

$$L_{\text{red}} = \frac{1}{2}|\dot{q}|^2 - V(q) - \frac{1}{2M^2} |\ddot{q} + \nabla V(q)|^2, \quad V(q) = \frac{g^2}{2} (x^2 y^2 + y^2 z^2 + z^2 x^2). \quad (2)$$

The volume factor L^3 multiplies the whole action and drops out of the equations of motion; the energies quoted below are energy densities, the total energy on the torus being $L^3 E$. Throughout, E is the energy of the Yang–Mills background. On the ghost-free manifold it is the total energy of the system, and the linearised ghost of (4) does not change it; the full Lee–Wick energy is indefinite, and in the nonlinear runs of Sec. 6 E refers to the Yang–Mills energy of the initial data, the ghost being seeded with initial energy fraction ε . The ghost amplitude is literally the violation of the ordinary Yang–Mills equation of motion. The rescaling $t \rightarrow t/M$, $q \rightarrow (M/g)q$ gives $L = (M^4/g^2)\tilde{L}$ with $g = M = 1$; henceforth time is in units of $1/M$, fields in units of M/g , and energy in units of M^4/g^2 . With the auxiliary field $w = \ddot{q} + \nabla V$ the dynamics is generated by the separable Hamiltonian

$$H = \frac{1}{2}|P|^2 - \frac{1}{2}|W|^2 + V(q) + w \cdot \nabla V(q) - \frac{1}{2}|w|^2, \quad q = p - w, \quad (3)$$

with $U(p, w) = V(q) + w \cdot \nabla V(q) - \frac{1}{2}|w|^2$, $\dot{p} = P$, $\dot{w} = -W$, $\dot{P} = -\partial_p U$, $\dot{W} = -\partial_w U$. The state is (p, w, P, W) , four 3-vectors. Two structural facts make the problem sharp:

- $w = W = 0$ is an invariant submanifold on which the dynamics is exactly Yang–Mills mechanics, $\ddot{p} = -\nabla V(p)$, whose maximal Lyapunov exponent scales exactly as $\lambda_{\text{YM}} \simeq 0.38 E^{1/4}$ by the absence of an intrinsic scale.
- Linearised about it, the normal sector decouples and the ghost obeys

$$\ddot{w} = -(1 + \text{Hess } V(p(t))) w, \quad (4)$$

a Hill-type equation with chaotic rather than periodic coefficients: a product of random matrices in the Furstenberg sense. Its largest Lyapunov exponent $\lambda_{\perp}(E)$ is the transverse Lyapunov exponent of the ghost-free manifold in the sense of the theory of riddled basins and blowout bifurcations [24, 25].

The eigenvalues of $\text{Hess } V$ satisfy $\lambda_{\min} \geq -\sqrt{2V}$ with equality on the ridges $x = \pm y$, $z = 0$ (and permutations), so the ghost mass matrix $1 + \text{Hess } V$ can become non-positive only for $E \geq \frac{1}{2}$ (a trajectory at rest on a ridge with $V = E$). Inside a channel ($|x| \gg |y|, |z|$) the negative eigenvalue is $\approx -3y^2$ and shrinks with depth; the tachyonic configurations lie near turning points on the ridges between channels, not deep inside them. Below $E = \frac{1}{2}$ the instability, if any, must be parametric.

3 Numerical methods

All integrations use a fourth-order Yoshida symplectic scheme [23] for the separable Hamiltonian (3) or its linearisation. Yang–Mills trajectories with small transverse action travel far along the channels, where the transverse frequency is $\sim |x|$; a fixed-step scheme becomes unstable at $|x| dt \gtrsim 1.57$, and channel depths are heavy-tailed (at $E = 1$ the maximum of $|p|$ over 400 time units has median 10, 99th percentile 80, and exceedance $\propto D^{-2}$), so every trajectory is sub-stepped individually whenever $\max_i |p_i| dt > 0.25$. Relative energy errors stay at $\sim 10^{-3}$ over $T = 7 \times 10^4$.

$\lambda_{\perp}$ is estimated from the slope of the accumulated log-growth of the renormalised ghost vector $(w, \dot{w})$ over the trailing half of the run, with an adaptive stopping rule on the ensemble standard error, the block-to-block drift and an $a + b/T$ trend extrapolation; the difference between the final estimate and the extrapolation ($\leq 2.5\%$ for $E \geq 0.015$, $\leq 10\%$ at $E = 0.008$) is quoted as the systematic uncertainty, the statistical standard error being much smaller. For the physical ghost

Table 1: Transverse Lyapunov exponents at the four lowest energies ($N = 2 \times 10^5$). The diagonal-ghost column is the single-mode rate (per-component estimator); the last column is its ratio to the parameter-free prediction (13), $0.0120 E^{9/4}$. Uncertainties are the gap between the final estimate and the $a + b/T$ extrapolation; statistical standard errors are 0.2–3%.

E	$\lambda_{\perp}$ (vector)	λ_{diag}	vector/diag	$\lambda_{\text{diag}}/0.0120 E^{9/4}$
0.03	9.92×10^{-6}	4.36×10^{-6}	2.27	0.971
0.02	4.02×10^{-6}	1.84×10^{-6}	2.18	1.021
0.015	2.13×10^{-6}	9.47×10^{-7}	2.25	1.002
0.008	$4.00 \times 10^{-7} (\pm 4\%)$	$2.29 \times 10^{-7} (\pm 10\%)$	1.75	0.995

the norm of the full six-component vector is renormalised, which is the standard estimator of the largest exponent of the coupled system. For the “diagonal” ghost of Sec. 4, in which the three components are independent scalar oscillators, each component is renormalised separately and the three log-growths are averaged: the exponent of the single-mode mechanism is the mean of three realisations of the same scalar process, not their maximum.¹ Ensembles of 10^3 (CPU) to 2×10^5 (GPU) trajectories were used; the GPU code integrates each trajectory inside a single kernel. Initial data are random phase points rescaled to the energy shell by the scaling symmetry, with a burn-in of 200 Yang–Mills times ($200 E^{-1/4}$; doubling or quartering it does not change the converged exponents). The instability is an ensemble-mean statement: at $E = 0.008$ the finite-time slopes of individual trajectories over $T \sim 10^5$ have a standard deviation several times the mean, and roughly 40% of them are negative, as expected for a wide finite-time Lyapunov distribution with a positive mean. Two implementations (numpy on CPU, CuPy on GPU) were used for the production runs. In addition, the channel statistics of Sec. 5, the Hessian spectrum, the transverse exponents for $E \geq 0.215$, the Floquet tongues of Sec. 7, the escape statistics of Sec. 6 and the full-system Lyapunov exponent were reproduced with a third, separately written integrator; where a number below comes from that re-run it is said so.

The nonlinear escape-time landscape (Sec. 6) uses the same integrator on the full system with escape declared at $\max|\text{component}| > 10^4$, about one time unit before the singularity; only the ordering of escape times between nearby initial conditions matters, so the same dt and threshold are used throughout. Growth laws of individual escaping trajectories were confirmed with an adaptive DOP853 integration at tolerance 10^{-11} . The code (numpy, C/OpenMP and CUDA implementations, the symbolic reductions, the analysis scripts and the data behind every figure and table) is at <https://github.com/foejoseph-lab/hdym>.

4 The transverse Lyapunov exponent

Figure 1 shows $\lambda_{\perp}(E)$. It is positive at every energy from $E = 1$ down to $E = 0.008$. Table 1 lists the four lowest energies, at which both the physical ghost and the diagonal ghost were measured with $N = 2 \times 10^5$ trajectories and run lengths $T = 1.2\text{--}1.8 \times 10^5$ (the same Yang–Mills ensembles for both). At $E = 0.015$ the physical value 2.13×10^{-6} agrees with three earlier independent runs ($2.08\text{--}2.35 \times 10^{-6}$; CPU $N = 1024$, $T = 5.4 \times 10^4$; GPU short and long). Positivity at the lowest energies does not rest on the transient part of the estimate: runs at $T = 2.6 \times 10^4$ and 5.4×10^4 agree within 10%, whereas a bounded ghost would give a ratio of 0.48, and the per-trajectory log-growth is linear in T over the whole trailing half of every run.

¹Renormalising the norm of the three-vector instead returns, at finite T , the largest of three finite-time exponents. At these run lengths the finite-time exponent distribution is wide (standard deviation 2–5 times the mean, 30–40% of individual trajectories with negative finite-time slope at $E \leq 0.03$), and the maximum of three overestimates the mean by a factor 1.5–1.7 over $E = 0.008\text{--}0.03$, decreasing to 1.1 at $E = 1$ as the finite-time distribution narrows. An earlier version of this work used that estimator and reported a corresponding 30–50% excess over Eq. (13).

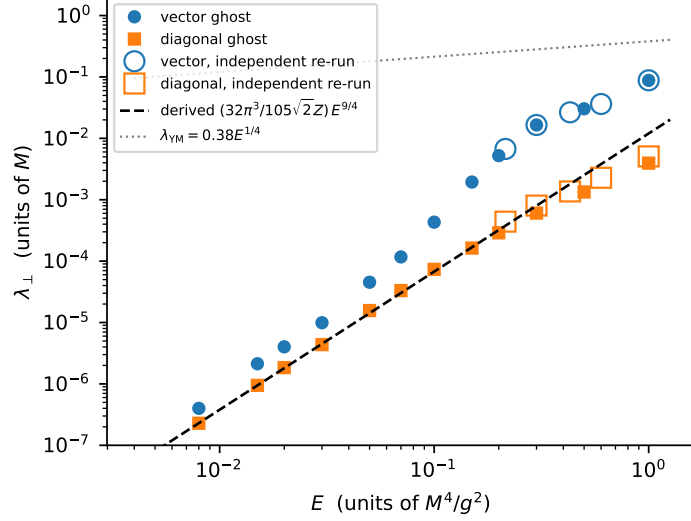


Figure 1: Transverse Lyapunov exponent of the ghost-free manifold. Circles: physical three-component ghost; squares: “diagonal” ghost in which the off-diagonal entries of Hess V are dropped (three independent scalar oscillators, per-component estimator of Sec. 3). Dashed: the parameter-free prediction of Sec. 5 for the diagonal ghost, $(32\pi^3/105\sqrt{2}) E^{9/4}/Z = 0.0120 E^{9/4}$ with $Z = 557.1$; nothing is fitted. Dotted: the Yang–Mills Lyapunov exponent for comparison. Filled symbols: GPU production runs ($N = 2 \times 10^5$ at $E \leq 0.03$, 10^5 above). Open symbols: the independent re-run of Sec. 3 at $E = 1, 0.6, 0.43, 0.3$ and 0.215 (512 trajectories each); its diagonal-ghost points use the three-vector norm estimator and lie above the per-component values by the 1.1–1.3 expected at these energies (footnote in Sec. 3).

Over $0.008 \leq E \leq 0.03$ the diagonal ghost follows

$$\lambda_{\text{diag}} = (0.0114 \pm 0.0010) E^{2.24 \pm 0.03}, \quad \text{equivalently} \quad \lambda_{\text{diag}} = (0.0120 \pm 0.0003) E^{9/4} \quad (5)$$

with the exponent fixed at $9/4$; the local slopes between successive points being 2.26, 2.32 and 2.13. The physical ghost lies a factor 2.2–2.3 above it at $E = 0.015$ – 0.03 and 1.75 at $E = 0.008$, with a local slope of 2.66 between the two lowest points; the excess $\lambda_{\perp} - \lambda_{\text{diag}}$ falls faster than $E^{9/4}$, so within the diagonal sector the asymptotic law for the physical three-component ghost is the diagonal one (the off-diagonal ghost components, which grow faster, are treated in Sec. 8). There is no threshold. Over the factor-of-four range of Table 1 an exponential law $\exp(-cE^{-1/4})$ with $c \simeq 3.5$ has a comparable local slope, so the fit alone does not exclude exponential suppression; that exclusion comes from the spectral analysis of Sec. 5, and the parameter-free agreement of the diagonal rate with (13) at four energies is the quantitative form of it.

Two further features of the curve are diagnostic. First, $\lambda_{\perp}(E)$ is not a single power law: it steepens between $E \simeq 0.1$ and 0.25 , flattens between 0.3 and 0.6 , and steepens again above 0.6 (in the re-run the local logarithmic slope is 2.7 between 0.215 and 0.3 , 1.3 and 0.9 between 0.3 and 0.6 , and 1.7 between 0.6 and 1). Second, the ratio between the physical vector ghost and the diagonal ghost (off-diagonal Hessian entries dropped) is 22 at $E = 1$, 27 at 0.3 , 18 at 0.2 , 12 at 0.15 , 5.9 at 0.1 , 2.9 at 0.05 , 2.2–2.3 at 0.015 – 0.03 and 1.75 at 0.008 (re-run, norm estimator for the diagonal ghost: 15–21 over $E = 0.215$ – 1). The reading is that above $E \sim 0.1$ the growth is dominated by mixing of the three degenerate ghost modes by the off-diagonal entries in the interior of the potential, while the low-energy power law is single-mode parametric driving by the diagonal entries; the steepening between 0.1 and 0.25 is the mixing contribution switching off. The re-steepening above 0.6 coincides with the exact tachyonic threshold $E = \frac{1}{2}$: the fraction of time with a non-positive ghost mass matrix vanishes identically below $E = \frac{1}{2}$ (measured zero at $E = 0.43$), is 0.13% at $E = 0.6$, and rises to 3% at $E = 1$.

5 Derivation of the $E^{9/4}$ law

5.1 Spectral formula and the u^{-6} tail

Because Yang–Mills mechanics is scale-free, the Hessian along a trajectory at energy E is $\text{Hess } V(t; E) = E^{1/2} h(E^{1/4} t)$ with h computed once at $E = 1$. Equation (4) is then a unit-frequency oscillator driven parametrically by a stationary process of strength $E^{1/2}$ and correlation time $E^{-1/4}$. For weak driving, the Lyapunov exponent of a randomly modulated oscillator $\ddot{w} + (1 + \xi(t))w = 0$ is [26, 27] $\lambda = \frac{1}{8} S_\xi(2)$, with $S_\xi(\omega) = \int \langle \xi(\tau) \xi(0) \rangle e^{i\omega\tau} d\tau$ the two-sided spectrum. Hence

$$\lambda_\perp(E) \simeq \frac{1}{8} E^{3/4} \hat{s}(2E^{-1/4}), \quad (6)$$

where $\hat{s}(u)$ is the spectrum of the relevant Hessian entry at $E = 1$. This is the classical statement that a fast oscillator coupled to a slow chaotic system responds to the chaotic spectrum at its own (doubled) frequency [28, 27]; the ghost is that result with the sign of the response reversed. It is also the Thouless/Born formula for the inverse localisation length of a one-dimensional Anderson problem in the time domain, with $\lambda_\perp$ the inverse localisation length and the Hessian the (correlated) disorder.

The measured spectrum of the diagonal entries at $E = 1$ has a power-law tail. A first estimate (512 trajectories, no ghost) gave $\hat{s} \propto u^{-5.6}$ to -6.2 over $u = 6$ –30 with a local exponential rate that keeps falling ($1.5 \rightarrow 0.23$); a high-statistics multitaper estimate (DPSS, $NW = 16$, $K = 8$; 2048 trajectories, $T = 2 \times 10^4$, 98 304 segments of 409.6 time units) of the mean diagonal entry gives local slopes -6.14 , -6.05 , -6.00 , -5.98 , -5.89 over $u = 5$ –8, 8–12, 12–20, 20–30, 30–45, and

$$u^6 \hat{s}(u) = A = 21.6 \pm 0.3 \quad (u = 8\text{--}45), \quad (7)$$

constant to $\pm 1\%$; beyond $u \simeq 45$ the ensemble runs out of sufficiently deep visits. It is not an exponential. Splitting the segments by their deepest excursion confirms the kinematic origin of the tail: segments whose maximum $|p|$ is below the median (10.0) reproduce the total spectrum for $u \lesssim 5$ and contribute nothing above $u \simeq 12$, while the deep half carries 60% of the power at $u = 12$, 92% at $u = 20$ and all of it for $u \geq 30$; a trajectory whose deepest excursion is D contributes nothing above $u \approx 2D$. (The off-diagonal entries have a u^{-2} tail, a different passage integral; the lowest Hessian eigenvalue has slopes -5.6 to -5.9 , like the diagonal entries.) Since $u = 2E^{-1/4}$, a u^{-6} tail maps onto $E^{3/4+6/4} = E^{9/4}$, and the predicted $\lambda_\perp$ reproduces the measured slope with no fitted parameter; using the measured $\hat{s}(u)$ itself in (6) predicts an effective exponent of 2.26 between $E = 10^{-2}$ and 10^{-4} ($u = 6.3$ –20). The direct measurements of Table 1 sit at $u = 4.8$ –6.7, at the lower edge of the range over which the u^{-6} plateau is established, so the exponent $9/4$ rests on the spectrum over $u = 8$ –45 as much as on the four measured points. For a bounded, analytic forcing of fixed bandwidth, adiabatic-invariance arguments give $\lambda_\perp \propto e^{-cE^{-1/4}}$ [29]. That is not the situation here: each trajectory’s forcing is analytic, but its instantaneous frequency $2|x|$ is unbounded along the channels, and the ensemble measure of reaching frequency u is a power law (Sec. 5.2), so the spectral tail is a power law. This is the basis for the statement that there is no exponential suppression at any energy.

5.2 Channel statistics

The tail can be derived. Consider the along-channel ghost mode w_x during a visit to the x -channel at depth $|x| = D$. It is driven by $K_{xx} = y^2 + z^2$; the transverse coordinates are adiabatic oscillators of frequency $|x|$ with actions J_y, J_z , so $y^2 = (J_y/D)(1 + \cos 2\phi_y)$: a cosine at frequency $2D$ with amplitude J_y/D . The two-sided spectrum per unit time of a slowly chirping cosine of amplitude a and instantaneous frequency Ω is $\pi \langle (a^2/2) \delta(u - \Omega) \rangle$, whence

$$\hat{s}_{xx}(u) = \frac{2\pi}{u^2} \left\langle (J_y^2 + J_z^2) \delta(u - 2|x|) \right\rangle_{\text{time}}. \quad (8)$$

Along the channel axis the potential vanishes, so at $E = 1$ the motion obeys $\frac{1}{2}\dot{x}^2 + J|x| = 1$ with $J = J_y + J_z$, the turning point is at $D_{\max} = 1/J$ (adiabatic relation, verified numerically to 0.1%; Fig. 2b), and a visit spends time $2dD/\sqrt{2(1-JD)}$ at depth D . If entries into the channel with transverse actions in $dJ_y dJ_z$ occur at rate $\nu_0 dJ_y dJ_z$, uniform in the actions at small J , then

$$\hat{s}_{xx}(u) = \frac{32\pi I \nu_0}{u^6}, \quad I = \int_{a+b<1} \frac{a^2 + b^2}{\sqrt{2(1-a-b)}} da db = \frac{64}{105\sqrt{2}}, \quad (9)$$

and inserting in (6),

$$\lambda_{\text{diag}}(E) = \frac{4\pi}{105\sqrt{2}} \nu_0 E^{9/4} = 0.0846 \nu_0 E^{9/4}. \quad (10)$$

The exponent decomposes as $\frac{3}{4}$ (kinematic) + $\frac{2}{4}$ (two transverse actions, whose uniform measure gives $P(J < j) \propto j^2$ and hence the D^{-2} exceedance of channel depths) + $\frac{4}{4}$ (the driving power $\propto y^4 \propto (J/D)^2$ with $J \lesssim 1/D$). No exponential can arise because the measure of small transverse actions is a power law.

Two remarks on what is and is not counted. The y - and z -channels also put power into K_{xx} at frequency $2|y|$ or $2|z|$ (through z^2 or y^2), and this power is present in the measured spectrum; but during those visits w_x has frequency $\sqrt{1+y^2} \approx |y|$, not 1, and the driving is detuned from its parametric resonance by $\sim 1/D$ against a resonance half-width $\lesssim 1/(2D^3)$. It does not drive growth. The same detuning is what prevents the off-diagonal entries, which carry far more power at $u \approx 5$, from driving the mixed modes inside channels. This can be checked quantitatively at the level of the tail constant, and the detuned share can be derived. Equation (9) gives $A_{\text{res}} = 32\pi I \nu_0 = 6.14$ for the resonant x -channel share of $\hat{s}_{xx}$. The remainder comes from the y - and z -channels, in two ways. During a y -channel visit the entry z^2 chirps directly, with amplitude J_z/D at frequency $2|y|$. In addition the along-channel coordinate is not slow at second order: the transverse oscillators exert the force $-\partial_y V = -y(x^2 + z^2) = -\text{sgn}(y) \sum_a J_a (1 + \cos 2\phi_a)$ on y , whose oscillating part produces the response $\delta y = \text{sgn}(y) \sum_a J_a \cos 2\phi_a / (4y^2)$ and hence a term $2y \delta y = (J_x \cos 2\phi_x + J_z \cos 2\phi_z) / (2D)$ in $K_{xx} = y^2 + z^2$, at the same frequency $2|y|$. The ϕ_z pieces are coherent, so the oscillating part of K_{xx} in the y -channel is $(3J_z/2D) \cos 2\phi_z + (J_x/2D) \cos 2\phi_x$, with power $(9J_z^2 + J_x^2)/4D^2$ once the independent phases are averaged; the z -channel contributes $(9J_y^2 + J_x^2)/4D^2$. The uniform measure on the two transverse actions makes all $\langle J_a^2 \rangle$ equal and the three channels equally visited, so resonant : detuned : total = 2 : 5 : 7,

$$A = \frac{7}{2} A_{\text{res}} = 21.5, \quad \frac{A_{\text{res}}}{A} = \frac{2}{7} = 0.286, \quad (11)$$

against the measured $A = 21.6 \pm 0.3$ of (7). The detuned 5/7 of the tail power sits at frequency $2|y|$ or $2|z|$ while w_x has frequency $\sqrt{1+y^2} \approx |y|$ or $\sqrt{1+z^2}$ during those visits, detuned from its parametric resonance by $\sim 1/D$ against a half-width $\lesssim 1/(2D^3)$; it drives nothing. If only the resonant share drives the ghost, the ratio of the measured diagonal-ghost exponent to the prediction of (6) applied to the *full* measured spectrum should be 2/7. It is 0.283, 0.285, 0.290, 0.276 at $E = 0.008, 0.015, 0.02, 0.03$ (Table 1; and 0.15 at $E = 0.05$, where $u = 4.2$ is below the onset of the u^{-6} regime). The tail is entirely channel power, its composition is derived, and (10), which counts only the resonant piece, is the whole story. The same accounting gives the tail constant of any entry of Hess V ; for the decoupled component studied in Sec. 5.4 below, driven by $K_{zz} = x^2 + y^2$, it predicts a tail $A_{zz}/A_{\text{res}} = \frac{9}{2}$ in units of the $n = 1$ resonant constant $A_{\text{res}} = (256\pi/15\sqrt{2})\nu_0$ (= 2.51 at $\mu_1 = 0.1$); measured $A_{zz}/A_{\text{res}} = 4.3$ –4.4 over $u = 8$ –16.

5.3 ν_0 from the density of states

If the flow is ergodic, ν_0 is a microcanonical flux and needs no dynamics. On the surface $|x| = A$, changing transverse variables to actions and angles (unit Jacobian) and using $\int dp_x p_x \delta(E_x -$

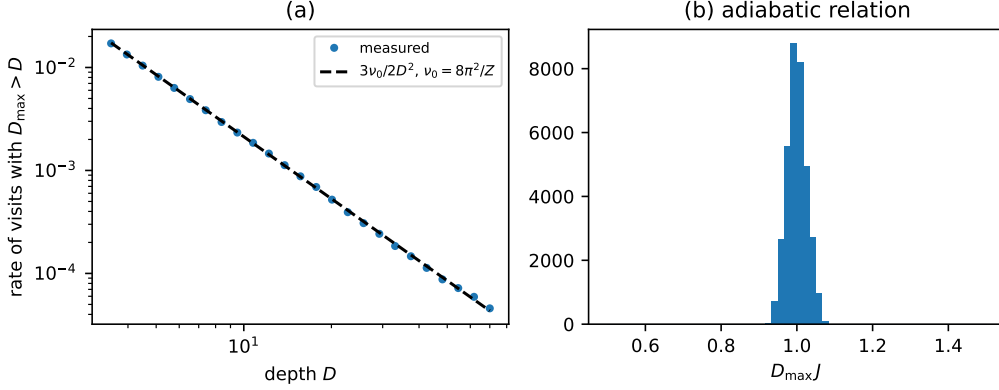


Figure 2: Channel statistics of pure Yang–Mills mechanics at $E = 1$ (512 trajectories, 2900 time units each, 35 585 channel visits). (a) Rate of visits to any channel reaching depth $D_{\max} > D$, against the parameter-free prediction $3\nu_0/2D^2$ with $\nu_0 = 8\pi^2/Z$. (b) The adiabatic relation $D_{\max}J = 1$ at entry.

$p_x^2/2 = 1$ for the along-channel flux, the entry-rate density is $(2\pi)^2/Z$ per channel end, independent of A , with two ends per channel:

$$\nu_0 = \frac{8\pi^2}{Z}, \quad Z = \int \delta(1 - H) d^3q d^3p = 4\pi\sqrt{2} \int \sqrt{1 - V(q)} \Theta(1 - V) d^3q. \quad (12)$$

Z is the classical density of states of Yang–Mills mechanics at $E = 1$, finite despite the open channels because they narrow as $1/|x|$ (the same integral underlies the discreteness of the quantum spectrum of the x^2y^2 potential [22]). Numerically $Z = 557.1$, so $\nu_0 = 0.1417$; the measured value from 35 585 channel visits is 0.146 over $D = 4$ –16 (Fig. 2a), with a slow softening to $\simeq 0.12$ at $D = 30$ –50. An independent re-run with a separately written integrator (34 676 visits) gives $\nu_0 = 0.140$ –0.146 over $D = 4$ –16, 0.134 at $D = 24$ –48, and $D_{\max}J = 0.999$ (median). A larger run (2048 trajectories, $T = 4000$, 194 602 visits) confirms each ingredient of the derivation separately: $J_{\text{peak}}/J_{\text{entry}}$ has median 1.000 (interquartile 0.983–1.017); $D_{\max}$ against $1/J_{\text{entry}}$ has log-log slope 1.00 and prefactor 1.01 for visits deeper than 6; the cumulative distribution of J_{entry} at small J has slope 2.05, i.e. a uniform action measure; and the exceedance of $D_{\max}$ has slope -2.06 . The deepest recorded visit reached $D = 105.6$ over 294.5 time units with $J_{\text{entry}} = 0.0095$ and a parabolic profile of the same J . The final, parameter-free result is

$$\lambda_{\text{diag}}(E) = \frac{32\pi^3}{105\sqrt{2}} \frac{E^{9/4}}{Z} \simeq 0.0120 E^{9/4} \quad (13)$$

against the measured diagonal-ghost values of Table 1: the ratio measured/predicted is 0.995, 1.002, 1.021, 0.971 at $E = 0.008, 0.015, 0.02, 0.03$ (mean 0.997, scatter 2%), and the free fit (5) gives exponent 2.24 ± 0.03 and, at fixed $9/4$, prefactor 0.0120 ± 0.0003 . Every number on the right-hand side of (13) is either a pure number or the density of states, and the agreement is at the level of the statistical and extrapolation uncertainties of the measurement. Above the range of Table 1 the diagonal rate is 1.10, 1.10, 1.09 times (13) at $E = 0.05, 0.07, 0.1$, where $u = 4.2$ –3.6 is at the onset of the u^{-6} regime and the measured spectrum lies slightly above the asymptotic power law, and it falls below (13) from $E \simeq 0.15$ on (0.97, 0.90, 0.76, 0.53, 0.33 at $E = 0.15, 0.2, 0.3, 0.5, 1$), where the resonance has moved into the knee of the spectrum. The single-mode law therefore holds to 10% over a factor 12 in energy, $0.008 \leq E \leq 0.1$. The resonant depth at these energies is $D = E^{-1/4} \simeq 2.4$ –3.3, at the lower edge of the range in which the channel asymptotics were established directly (the D^{-2} plateau of Fig. 2a begins at $D \simeq 4$); that the law already holds there says the adiabatic description of a visit is accurate from shallower depths than the plateau

Table 2: Two-field model: diagonal-ghost exponent against the parameter-free prediction (14) with $n = 1$ and the regulator factors ($N = 10^5$ at $E = 0.1$, 2×10^5 otherwise). λ_z is the exponent of the decoupled third component w_z , driven by $K_{zz} = x^2 + y^2$, all of whose channel power is detuned.

E	μ_1	λ_{diag}	prediction	ratio	$\lambda_z/\lambda_{\text{diag}}$
0.1	0.1	8.08×10^{-5}	9.32×10^{-5}	0.87	0.235
0.03	0.1	7.76×10^{-6}	8.15×10^{-6}	0.95	0.017
0.01	0.1	8.85×10^{-7}	8.60×10^{-7}	1.03	0.005
0.03	0.03	5.75×10^{-6}	6.04×10^{-6}	0.95	0.015

test requires. The apparent softening of ν_0 to $\simeq 0.12$ – 0.13 at $D \gtrsim 20$ is at the depths where the ensembles run out of visits (a $D = 50$ visit lasts ~ 140 time units against segments of 410) and we do not build on it; if the microcanonical flux is exact, as the derivation assumes, there is nothing to soften. The remaining factor between the physical vector ghost and the diagonal ghost (2.2–2.3 at $E = 0.015$ – 0.03 , 1.75 at $E = 0.008$) is not part of the single-mode argument and is attributed to interior off-diagonal mixing, where the three modes are degenerate and no detuning protects them; it falls with energy, and within the diagonal sector the asymptotic law for the physical ghost is (13) itself.

5.4 The exponent as a count of transverse actions; test on the two-field model

Nothing in the derivation is specific to three degrees of freedom. For a diagonal potential $V = \frac{1}{2} \sum_{i < j} q_i^2 q_j^2$ with $N = n + 1$ degrees of freedom, so that a channel has n transverse actions, the microcanonical entry flux is $\nu_0 = 2(2\pi)^n/Z_N$ per channel, the passage integral is the Dirichlet integral $I_n = \int_{\sum a < 1} (\sum a_i^2) [2(1 - \sum a)]^{-1/2} d^n a = \sqrt{2\pi} n / \Gamma(n + \frac{5}{2})$, and

$$\lambda_{\text{diag}} = \frac{\pi}{16} I_n \nu_0 E^{(n+7)/4} = \frac{2^{n-5/2} \pi^{n+3/2} n}{\Gamma(n + \frac{5}{2})} \frac{E^{(n+7)/4}}{Z_N}, \quad \frac{A_{\text{res}}}{A} = \frac{4}{9n - 4}, \quad (14)$$

which reduces to (13) and (11) at $n = 2$. The exponent decomposes as $\frac{3}{4}$ (kinematic) + $\frac{n}{4}$ (the measure of n small actions is $\propto J^n$) + 1 (driving power $\propto (J/D)^2$); the middle term is the only one that knows N . The $n = 1$ case is the two-field $x^2 y^2$ model, which is the invariant plane $z = 0$ of the present system, i.e. the ansatz $A_i^a = \text{diag}(x, y, 0)$, with prediction $\lambda \propto E^2$ and prefactor $2\sqrt{2}\pi^2/15 = 1.861$ over Z_2 , and a predicted resonant share of $4/5$.

The two-field model needs one modification: in two dimensions the channel cross-section falls as $1/|x|$ rather than $1/x^2$, so the classical density of states is logarithmically divergent [22], the mean channel-visit time $\int dJ/J$ diverges, and every time average, including λ_{YM} , decays as $1/\ln T$. We close the channels with a mass, $V = \frac{1}{2}x^2 y^2 + \frac{1}{2}\mu^2(x^2 + y^2)$ with $\mu^2 = \mu_1^2 E^{1/2}$, which keeps the model exactly scale-free (every energy is the rescaled $E = 1$ system with regulator μ_1) and makes $Z_2(\mu_1) = 8\sqrt{2}\pi\sqrt{1+b^2} [K(m) - E(m)]$, $b^2 = \mu_1^4/2$, $m = 1/(1+b^2)$, finite: $Z_2 = 189.7$ at $\mu_1 = 0.1$ and 275.3 at $\mu_1 = 0.03$. The regulator enters the passage integral through the turning-point condition $JD + \frac{1}{2}\mu^2 D^2 = 1$, which multiplies the prediction by $(1 - \mu_1^2/2\sqrt{E})^{5/2}$ (0.88 at $E = 0.01$, $\mu_1 = 0.1$), and shifts the ghost mass to $1 + \mu^2$, a 1% effect; the law applies for $E \gg \mu_1^4/4$, i.e. while the resonant depth lies well inside the channel cutoff (a fixed physical mass μ would instead produce a crossover at $E \sim \mu^4$, which is a property of the regulator, not of the ghost). Every ingredient was checked separately at $E = 1$: the flux $\nu_0 = 4\pi/Z_2$ is reproduced to 0.3% at both μ_1 by 6×10^4 channel visits, the entry measure is uniform in J , $DJ + \frac{1}{2}\mu^2 D^2 = 1.000$ at the turning point, and the measured tail $u^5 \hat{s}_{xx}$ is $(1.25 \pm 0.02) \times A_{\text{res}}(1 - \mu^2 D^2/2)^{5/2}$, flat over $u = 8$ – 45 , the $5/4$ being the $n = 1$ value of (14).

Table 2 gives the result. The local slope between $E = 0.03$ and 0.01 is 1.98 against the predicted 2.05 (including the energy dependence of the regulator factor), and the prefactor is

reproduced to 5% at $u = 4.8$ and 3% at $u = 6.3$, the same agreement the three-field data show at the same u . Changing μ_1 from 0.1 to 0.03 changes Z_2 by 45% and the regulator correction by 7%; the ratio to prediction is 0.951 in both cases, which tests the $1/Z$ factor in a way the three-field model cannot. With the measured spectrum in (6) the ghost sees 0.83 of the tail power at $E = 0.03$ and 0.01, against the predicted $4/5$. Finally $\lambda_z/\lambda_{\text{diag}}$ falls by a factor 47 between $E = 0.1$ and 0.01 (0.235, 0.017, 0.005): detuned channel power is inert, as assumed throughout. The decomposition of the exponent is therefore tested at $n = 1$ and $n = 2$, and the single number left in (14) that is not a closed form is the density of states $Z_3 = 557.1$. Within the adiabatic and weak-driving approximations the law is exact given ergodicity of the Yang–Mills flow on the energy shell, with corrections $O(E^{1/2})$; the 5% at $u = 4.8$ in both models is what that looks like.

6 Nonlinear escape

6.1 Mechanism and time scale

Full nonlinear evolution on a grid of (E, ε) , with ε the initial ghost energy fraction, shows escape to a finite-time singularity in every cell in which escape was resolved ($E \gtrsim 0.06$ at $\varepsilon = 10^{-4}$; lower cells were censored by the run length). Escape times satisfy $t_{\text{esc}}\lambda_{\perp} \approx 2\text{--}5 \approx \ln(q_{\text{typ}}/w_0)$: the linear instability sets the clock, and the ε -dependence follows $\frac{1}{2}\ln(1/\varepsilon)$ as expected: the measured shift in $t_{\text{esc}}\lambda_{\perp}$ between $\varepsilon = 10^{-4}$ and 10^{-2} is 2.3, against $\frac{1}{2}\ln 10^2 = 2.30$; the re-run at $E = 0.3$ (median $t_{\text{esc}} = 98.5$ and 222.6 at the two ε , $\lambda_{\perp} = 0.0165$) gives 2.05. (The corresponding E -dependence, $\frac{1}{4}\ln(1/E)$, has not been checked.) Adaptive re-integration of escaping trajectories shows that the blow-up is not a clean power law in the state norm: $|w|, |P|, |W|$ grow monotonically with local per-decade exponents fluctuating between ~ 1 and ~ 10 while $|q|$ is non-monotonic. Sampling the last stretch before t_* reveals the mechanism: q is thrown along a channel axis with a transverse displacement ρ for which $1 + \lambda_{\min}(\text{Hess } V) = 1 - 3\rho^2 \ll 0$ (measured -49.5 at $\rho = 4.11$, -160.5 at $\rho = 7.34$); the hugely negative ghost mass matrix drives w , and large w drives q further out. The runaway is a self-accelerating tachyonic loop. The re-run sees the same sequence in ensemble medians at $E = 0.3$: as the largest component passes $10, 10^2, 10^3$, ρ grows $1.0 \rightarrow 1.4 \rightarrow 1.9$, $1 + \lambda_{\min}$ falls $-1 \rightarrow -4 \rightarrow -9$ (tenth percentile -35) and $|w|$ grows $3 \rightarrow 22 \rightarrow 110$. In pure Yang–Mills the transverse displacement shrinks with depth adiabatically, which is why the tachyonic fraction vanishes below $E = \frac{1}{2}$; in the nonlinear runaway the ghost pumps it up. In Smilga’s terminology [15] the ghost is malicious at every energy at which the escape has been resolved ($E \gtrsim 0.06$); below that the linear instability is established but the run lengths needed to follow the escape exceed those used here. No polynomial first integral independent of H exists up to degree 4 in the twelve phase-space variables (numerical null-space search of $\{H, F\} = 0$ over 1820 monomials, which recovers exactly the three partial energies for a separable control potential), so none of the known global-stability mechanisms [13, 16] is available at that order.

6.2 The escape-time landscape

Since every trajectory escapes, the geometric object is the escape-time function t_{esc} (initial condition) and the level sets $\{t_{\text{esc}} < t\}$. Figure 3 shows t_{esc} on two-dimensional sections at $E = 0.3$, $\varepsilon = 0.01$. Both sections show stretched and folded bands at large scale, the imprint of the chaotic base flow, on top of a texture that is rough at the grid scale: neighbouring cells differ in t_{esc} by a median of 30 time units against a median t_{esc} of 95, and the lag-1 correlation of $\log t_{\text{esc}}$ is 0.2–0.4. The ghost section is not smoother in this respect even though the Yang–Mills initial condition is identical in every cell, because the ghost back-reacts on q at $O(w)$ from the first step and the base flow amplifies the difference at λ_{YM} ; the ghost phase contributes only a $\pm 10\%$ smooth modulation of the median escape time.

The quantitative statement is the uncertainty exponent [30, 31]: for pairs of initial conditions at separation δ , the fraction $f(\delta)$ whose escape times differ by more than a threshold scales as δ^α ,

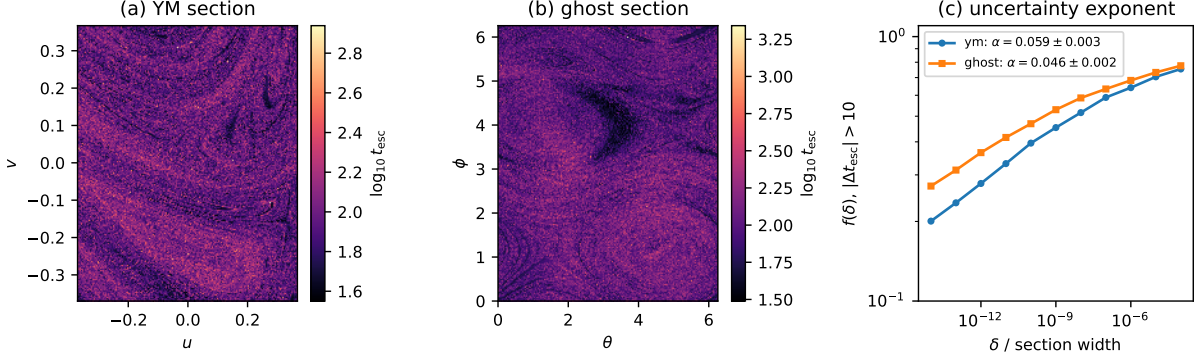


Figure 3: Escape-time landscape at $E = 0.3$, $\varepsilon = 0.01$. (a) 256^2 Yang–Mills section: the ghost is fixed and the Yang–Mills initial condition is varied over a patch of side $E^{1/4}$ in the (x, y) plane. (b) 256^2 ghost section: the Yang–Mills point is fixed and the ghost orientation (θ, ϕ) is varied over the torus. (c) Uncertainty-exponent test: the fraction $f(\delta)$ of pairs of initial conditions at separation δ whose escape times differ by more than 10 time units, 2×10^4 pairs per δ , on both sections.

and the level-set boundaries have dimension $d = 2 - \alpha$ on the section. Figure 3c shows $f(\delta)$ over eleven decades, from 10^{-4} to 10^{-14} of the section width, with no flattening at the bottom:

$$\alpha_{\text{YM}} = 0.059 \pm 0.003, \quad \alpha_{\text{ghost}} = 0.046 \pm 0.002 \quad (|\Delta t_{\text{esc}}| > 10), \quad (15)$$

and 0.04–0.07 for thresholds between 5 and 20 time units. The boundaries of the escape level sets are fractal with dimension 1.95 ± 0.01 on both sections, essentially area-filling. The standard chaotic-scattering estimate $\alpha = \kappa/\lambda$ [31], with $\kappa = 0.027$ the escape rate fitted to the exponential tail of the t_{esc} distribution, gives $\alpha \simeq 0.10$ with $\lambda = \lambda_{\text{YM}}(0.3) = 0.28$; the measured value implies an effective separation rate $\simeq 0.5$. A Benettin measurement on the full twelve-dimensional flow at $\varepsilon = 0.01$ (400 trajectories, renormalisation every time unit, the last 10 time units before escape excluded) gives $\lambda_{\text{max}} = 0.33 \pm 0.01$ as a per-trajectory time average, against 0.29 for pure Yang–Mills in the same code: the finite ghost raises the exponent only modestly, to $\alpha \simeq 0.08$. The ensemble-weighted rate $\sum \ln(\delta/\delta_0)/\sum t = 0.42$, which is dominated by the fastest-separating trajectories, gives $\alpha \simeq 0.06$, within the measured range. The small α is thus mainly a generalised-Lyapunov effect, the escape statistics being controlled by the tail of the finite-time exponent distribution rather than its mean. Either way, the conclusion is that the set of initial conditions that remain close to the ghost-free manifold for any prescribed time has no interior: the manifold is a repeller whose “basin” is a fractal of full dimension, at every scale down to the arithmetic.

7 Invariant sets on which the ghost is bounded

In a Hamiltonian system nothing attracts, so the question “where can a ghost sit” is a question about invariant sets on which the linearised ghost is bounded. On a periodic orbit of the base flow Eq. (4) is a genuine Hill equation. Figure 4 shows the result on the diagonal orbit: the ghost is Floquet-stable over most of the energy range, with narrow instability tongues at $E \in (0.165, 0.222)$ (along-diagonal mode), $(0.25, 0.585)$ and $E > 2.9$ (ridge mode), and $(0.117, 0.127)$, $E > 7.2$ (z mode). Two points deserve emphasis. First, for $E \in (0.585, 2.9)$ the ghost mass matrix is negative on part of every period yet the ghost is Floquet-stable: a transiently tachyonic mass is not an instability on a periodic orbit. Second, at $E = 0.3$ the ridge mode on this orbit grows at rate 0.11–0.13, six to ten times the chaotic-average $\lambda_{\perp}$; this orbit is the ridge on which the tachyonic configurations of Sec. 2 live, and the lower edge of its tongue, $E = 0.25$, coincides with the end of the steepening of $\lambda_{\perp}(E)$. Whether this is more than a coincidence requires the same analysis on

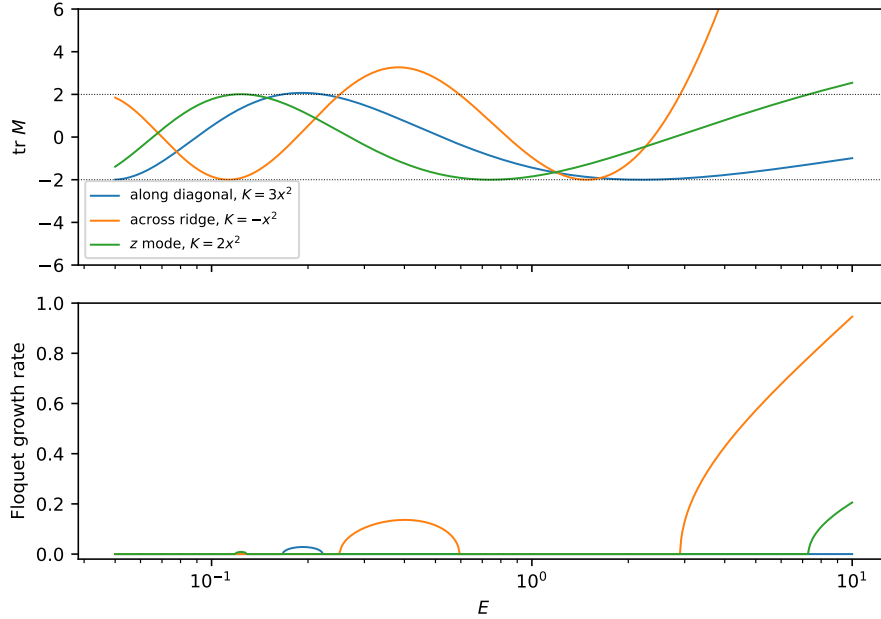


Figure 4: Floquet stability of the linearised ghost on the diagonal periodic orbit $x = y$, $z = 0$ of Yang–Mills mechanics ($\ddot{x} = -x^3$). The three ghost modes decouple with constant eigenvectors and $K = 3x^2$ (along the diagonal), $-x^2$ (across the ridge, the direction whose eigenvalue reaches the bound $-\sqrt{2V}$), $2x^2$ (z). Top: trace of the monodromy matrix; bottom: Floquet growth rate.

the other short periodic orbits (the channel and “circular” orbits), which we have not carried out. All such orbits are hyperbolic within the base flow, so these are landing points for the ghost but not for the trajectory: saddles within a saddle, of measure zero twice over. The only candidate for a positive-measure set on which the ghost stays bounded is the regular islands of Yang–Mills mechanics [21], on which the driving is quasi-periodic; whether any survive in the three-component system with a ghost attached is open.

8 The off-diagonal ghost sector

The colour-diagonal ansatz is a consistent truncation, but the Lee–Wick ghost of the full homogeneous theory has nine components, of which the ansatz keeps three. Linearising the full homogeneous system (all nine A_i^a , $A_0 = 0$) about a diagonal ghost-free solution, the six off-diagonal components u_i^a ($a \neq i$) split exactly into three equivalent blocks (u_{12}, u_{21}) , (u_{13}, u_{31}) , (u_{23}, u_{32}) , with quadratic Lagrangian

$$L_2 = \frac{1}{2}|\dot{u}|^2 - \frac{1}{2}u^T m u - \frac{1}{2}|\ddot{u} + m u|^2 + \frac{1}{2}j^2, \quad m = \begin{pmatrix} z^2 & -xy \\ -xy & z^2 \end{pmatrix}, \quad j = x\dot{u}_{12} - y\dot{u}_{21} - \dot{x}u_{12} + \dot{y}u_{21}, \quad (16)$$

for the first block. The first three terms are the structure of (2) with m the corresponding block of the Hessian of the nine-dimensional Yang–Mills potential; the last is the $(D^\mu F_{\mu 0})^2$ piece of the Lee–Wick action, j being the linearised colour charge $(A_i \times \dot{A}_i)_3$. It vanishes identically on the ansatz and is why (2) has no such term. Each block carries one exact gauge zero mode $u = \theta \times A$, one conserved linearised Gauss charge, and three physical degrees of freedom: one of Yang–Mills type and two ghosts, the second ghost being the longitudinal polarisation of the massive vector, which has no diagonal counterpart (in a channel the two sit at frequencies 1 and $\sqrt{1+x^2}$, the j^2 term lifting the second). The count is $6 + 9$ for the full homogeneous theory, as it must be.

Two facts follow. First, the off-diagonal Yang–Mills directions are neutral. The singular-value decomposition $A_i^a = (O_c \text{diag}(x, y, z) O_s)_i^a$ identifies them as the three colour rotations, which are

Table 3: Off-diagonal ghost exponents $g_1 \geq g_2$ of one block of (16), statistical errors $\leq 0.3\%$, against the diagonal single-mode law (13).

E	g_1	g_2	g_1/g_2	$g_1/(0.0120E^{9/4})$	local slope of g_1
1	1.180×10^{-2}	5.88×10^{-3}	2.01	0.98	
0.3	3.371×10^{-3}	1.73×10^{-3}	1.95	4.2	1.04
0.1	6.155×10^{-4}	3.94×10^{-4}	1.56	9.1	1.55
0.03	2.625×10^{-4}	1.46×10^{-4}	1.80	58	0.71
0.01	1.326×10^{-4}	7.75×10^{-5}	1.71	349	0.62

gauge, and the three spatial rotations, whose generators J_{ij} are conserved; the diagonal ansatz is the $J = 0$ sector of homogeneous Yang–Mills mechanics, not a measure-zero slice of a larger chaotic system. The linearised Yang–Mills block has two conserved linear functionals and a zero mode, forcing all four of its exponents to vanish; numerically they are $\pm \ln T/T$. The homogeneous theory therefore has no “generic” background beyond the one studied here, and the fragmentation referred to in Sec. 9 is an inhomogeneous effect only.

Second, the off-diagonal ghost grows faster than the diagonal one, and with a different law. Table 3 gives the two positive Lyapunov exponents of the block along Yang–Mills backgrounds at five energies ($N = 2 \times 10^4$ – 2×10^5 trajectories, two vectors with Gram–Schmidt, same estimator as Sec. 3).

The exponent is not a single power law: the local slope falls from 1.5 to 0.6 between $E = 0.3$ and 0.01, far below the $9/4$ of the diagonal sector and below the $5/4$ that the u^{-2} tail of the off-diagonal Hessian entry xy would give if it drove a degenerate ghost pair resonantly. At $E = 0.01$ the off-diagonal ghost grows 350 times faster than the diagonal one, and the ratio increases roughly as $E^{-1.6}$. The mechanism can be isolated by switching off the $(D^\mu F_{\mu 0})^2$ term ($c_j = 0$ in the code). Then $m = z^2 \mathbf{1} - xy \sigma_x$ has the fixed eigenvectors $(1, \pm 1)$ and the pair decouples exactly into two scalar Hill equations $\ddot{w}_\pm = -(1 + z^2 \mp xy) w_\pm$. In the x -channel the entry $xy = \sqrt{2J_y|x|} \cos \phi_y$ chirps at frequency $|x|$, not $2|x|$, with an amplitude that is not suppressed by depth, so it meets the parametric resonance at $|x| = 2$; and because without the j^2 term both $w_\pm$ keep frequency $\simeq 1$ in every channel, the y -channel resonates as well. The passage integral of Sec. 5.2 with one power of J gives $u^2 \hat{s}_{xy} = 16\sqrt{2}\pi\nu_0/15 = 0.672$ (measured 0.68–0.72 over $u = 4$ –20 along $E = 1$ trajectories) and

$$\lambda_\pm = \frac{4\sqrt{2}\pi^3}{15} \frac{E^{5/4}}{Z} = 0.0210 E^{5/4}, \quad (17)$$

the u^{-2} tail noted in Sec. 5.2 turned into a rate. Measured with $c_j = 0$: $g_1 = g_2$ to 5%, 5.07×10^{-3} at $E = 0.3$ and 1.47×10^{-3} at $E = 0.1$ against 4.66×10^{-3} and 1.18×10^{-3} , with 75% of the growth at $E = 0.1$ in the 27% of the time spent in channels; the excess is the knee of the xy spectrum at $u \leq 3.6$ and a modulation depth $\sqrt{2E}$ that is not yet small. In the actual theory the j^2 term lifts the longitudinal ghost to frequency $\sqrt{1 + x^2}$ inside a channel and detunes this resonance, which is why g_1 is 2.5 times smaller than (17) at $E = 0.1$ and $g_1 \neq g_2$. The growth nevertheless stays in the channels: at $E = 0.1$, 79% of the log-growth of the leading vector accrues in the 27% of the time spent at depth $D > 3$, the same split as with $c_j = 0$. What changes with the energy is the character of the events. Figure 5 shows $Q = \sigma^2 T/\lambda$, the variance of the per-trajectory finite-time exponent times the run length over its mean. For growth that arrives in discrete passages Q is the typical log-gain per passage and $\lambda T/Q$ the number of passages per trajectory, independent of T ; the reading requires $\lambda T \gg 1$, since at small λT the variance is dominated by bounded oscillation of the log-amplitude, which scales as $1/T$ rather than diffusively, and only runs with $\lambda T \geq 1.5$ are used (this excludes the low-energy diagonal and two-field runs, whose λT is 0.03–0.5). The parametric mechanisms give Q of order one, flat in E : 0.87–1.05 for the diagonal ghost at $E = 0.3$ –1 and 2.7–3.6 for the $c_j = 0$ block, whose kicks are larger because the xy chirp is not depth-suppressed. The off-diagonal ghost of the actual theory climbs as $Q \simeq 1.9E^{-1/2}$ from

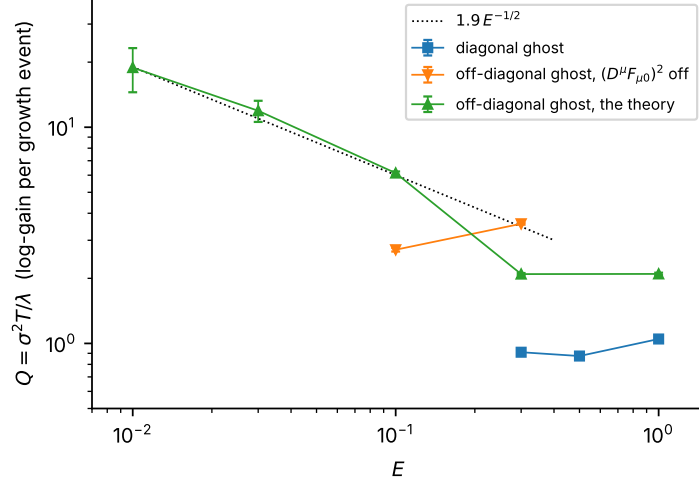


Figure 5: Log-gain per growth event, $Q = \sigma^2 T / \lambda$, from the per-trajectory finite-time exponents of the production runs with $\lambda T \geq 1.5$ e-folds (bootstrap errors). The parametric mechanisms (diagonal ghost; off-diagonal block with the Gauss-charge term off) give Q of order one, independent of E ; the off-diagonal ghost of the actual theory climbs as $E^{-1/2}$ (dotted), to one or two growth events per trajectory at $E = 0.01$.

$E = 0.3$ down to $Q = 19 \pm 4$ at $E = 0.01$ (bootstrap error over trajectories), where $\lambda T / Q = 1.7$: the 32 e-folds of a typical trajectory come from one or two deep channel visits, during each of which the ghost gains e^{19} . Fewer than 0.2% of trajectories have a negative finite-time exponent, far fewer than the $\sim 20\%$ that a Poisson count of 1.7 events would leave event-free, so the rare large events sit on top of a diffuse positive drift, presumably the detuned remnant of the channel resonance. The large events are not the small kick per passage of parametric resonance but an instability that persists for the duration of a visit, and the shallow energy dependence of Table 3 is its measure of deep visits. The only term of (16) whose frozen coefficients change sign inside a channel is the Gauss-charge term, which is where a derivation would start; we leave it to a sequel (Sec. 9).

The consequence for the headline result is a strengthening with a change of scope: (13) is the law of the colour-diagonal ghost, and the transverse exponent of the ghost-free manifold of the full homogeneous theory is set by the off-diagonal sector, which is less benign at every energy below $E \simeq 1$ and increasingly so as $E \rightarrow 0$. There is still no threshold and no exponential suppression.

9 Discussion

Scope. Instability of an exact sector implies instability of the full classical Lee–Wick Yang–Mills theory on T^3 , as an existence statement. The torus is also what makes the homogeneous sector physically meaningful rather than a formal truncation: the inhomogeneous modes carry momenta $\geq 2\pi/L$, and on a small torus (L below the inverse field strength) they are gapped and the zero-momentum sector dominates the low-energy dynamics [20]. Within the homogeneous sector the diagonal ansatz is neutral (Sec. 8); on a large torus the homogeneous condensate is unstable to inhomogeneous modes at rates of order λ_{YM} , which at low energy vastly exceeds $\lambda_{\perp}$; a real homogeneous configuration then fragments and thermalises long before its ghost grows, and the relevant follow-up is ghost growth in a thermal-like Yang–Mills state, for which the lattice machinery of glasma-instability studies exists [32, 33]. Nothing here addresses the quantum theory directly.

The fakeon classical limit. In the fakeon prescription the classical limit is defined by solving the ghost equation with the half-retarded, half-advanced Green function and substituting back, which discards precisely the homogeneous solutions of (4) [3]. The ghost-free manifold $w = W = 0$ is that projection at leading order. Our results say that this manifold is a repeller at every energy, with polynomially short growth times, and that its “basin” has no interior. The projection is therefore a definition rather than a dynamical statement: the classical projected theory is not the limit of nearby classical solutions, and any perturbation at the level of 10^{-12} of the initial data lands in the fractal. This does not refute the prescription, which is a statement about the quantum theory and need not have a classical limit in this sense, but it identifies the question a consistent classical limit must answer: what principle keeps the system on a measure-zero repelling manifold.

Contrast with benign and metastable ghosts. The globally stable and benign ghost systems in the literature [13, 14, 15] have interactions chosen for stability, and several are integrable or nearly so. Gross, Strumia, Teresi and Zirilli [10] argued that a weakly coupled ghost undergoes “spontaneous lockdown” rather than run-away as long as it is non-resonant, and that resonances are what make it behave statistically and run away, and they estimated run-away rates, including for four-derivative gravity. The present system is a case study in the resonant branch of that picture with the resonance structure supplied by a chaotic base flow rather than assumed: the interactions are fixed by gauge symmetry, the frequencies at which the ghost is driven form a continuum with a power-law tail, and the rate is a parameter-free function of the energy. The recent numerical study of ghost–normal scalar fields in [11], which finds instability mediated by spectral energy transfer with transient metastable windows for high-frequency, low-amplitude data, is the field-theoretic counterpart. We conjecture that a ghost coupled to a chaotic normal sector is never benign: the argument of Sec. 5 needs only a chaotic base flow whose phase-space measure near the resonant configurations is a power law, which is generic. This is testable in the existing toy models by adding a chaotic term. Salvio’s “Metastability in quadratic gravity” [9] finds that runaways are avoided at energies well below a threshold set by the couplings, and that the threshold is high enough to cover the cosmological applications. The present result refines rather than contradicts that conclusion: below the crossover $E \sim \frac{1}{2}$ the ghost here is not protected by any threshold, but its growth rate is the polynomially small (13), so that for the energies of the cosmological paragraph below it is metastable in practice, as he concludes, while not exponentially so. What this paper adds is the mechanism and the rate in the regime in which the normal sector is chaotic.

Cosmological scaling. The natural energy unit M^4/g^2 translates, for the gravitational case, to $E \sim \rho/(m_\chi^2 M_P^2) \sim H^2/m_\chi^2$ with m_χ the ghost mass: the Einstein–Hilbert normalisation M_P^2 plays the role of $1/g^2$ and the ghost mass that of M , so this is dimensional matching, not a derived correspondence. Today, for a ghost mass within a few orders of the Planck mass, $E \sim 10^{-120}$ and (13) gives a growth rate of order $10^{-270}m_\chi$ for the diagonal sector; the off-diagonal sector of Sec. 8, whose law is not derived, gives a rough bound of order $10^{-75}m_\chi$ if its measured slope $\simeq 0.6$ is extrapolated. Either way the absence of exponential suppression is irrelevant in practice, and the classical ghost is harmless at all post-inflationary energies. The crossover $E \sim \frac{1}{2}$ corresponds to $H \sim m_\chi$, i.e. to inflation if the ghost mass is comparable to the Hubble rate then, which is the regime in which the fakeon consistency bound $m_\chi \gtrsim m_\phi/4$ of [4] must hold. The classical tachyonic threshold and the quantum consistency bound are the same line seen from two sides; the passage through it at the end of inflation is the one epoch at which the nonlinear dynamics studied here could have mattered.

Open items. (i) The vector/diagonal factor ($\simeq 2.2$ at $E = 0.015$ – 0.03 , 1.75 at $E = 0.008$): a derivation of the mode-mixing contribution and of the rate at which it dies off, which would turn the asymptotic statement into a prediction for the physical ghost at all $E \lesssim 0.03$. (ii) The finite-time Lyapunov distribution of the full flow, whose tail sets α (Sec. 6). (iii) The regular

islands as a possible positive-measure refuge. (iv) The rate law of the off-diagonal ghost sector (Sec. 8), whose derivation is the natural sequel to this paper. (v) Bianchi IX minisuperspace of quadratic gravity, in which the same calculation can be done for the theory of actual interest, with the mixmaster billiard of Damour, Henneaux and Nicolai [12] as the base flow and the Weyl ghost as the transverse direction; the lesson of Sec. 8, that constraint terms which vanish on an ansatz can couple the normal and ghost sectors off it, applies there to the Hamiltonian and momentum constraints.

Acknowledgements

The numerical code, the derivations of Sec. 5, and much of the analysis in this paper were developed in interactive sessions with Claude (Anthropic). The choice of problem, the numerical experiments, and the judgement calls throughout are the author’s, as is responsibility for any errors.

References

- [1] K. S. Stelle, Phys. Rev. D **16**, 953 (1977).
- [2] D. Anselmi, JHEP **06**, 086 (2017), arXiv:1704.07728.
- [3] D. Anselmi, JHEP **02**, 141 (2018), arXiv:1801.00915; Class. Quantum Grav. **36**, 065010 (2019), arXiv:1809.05037.
- [4] D. Anselmi, E. Bianchi and M. Piva, JHEP **07**, 211 (2020), arXiv:2005.10293.
- [5] T. D. Lee and G. C. Wick, Nucl. Phys. B **9**, 209 (1969); Phys. Rev. D **2**, 1033 (1970).
- [6] B. Grinstein, D. O’Connell and M. B. Wise, Phys. Rev. D **77**, 025012 (2008).
- [7] J. F. Donoghue and G. Menezes, Phys. Rev. D **100**, 105006 (2019), arXiv:1908.02416; Phys. Rev. Lett. **123**, 171601 (2019), arXiv:1908.04170.
- [8] A. Salvio and A. Strumia, JHEP **06**, 080 (2014); A. Salvio, Front. Phys. **6**, 77 (2018).
- [9] A. Salvio, “Metastability in quadratic gravity,” Phys. Rev. D **99**, 103507 (2019), arXiv:1902.09557.
- [10] C. Gross, A. Strumia, D. Teresi and M. Zirilli, “Is negative kinetic energy metastable?,” Phys. Rev. D **103**, 115025 (2021), arXiv:2007.05541.
- [11] J. Wysong, S. Overvaag, H. Lim and J.-H. Kimn, “Numerical investigations of stable dynamics in the presence of ghosts,” arXiv:2604.25635.
- [12] T. Damour, M. Henneaux and H. Nicolai, Class. Quantum Grav. **20**, R145 (2003), arXiv:hep-th/0212256.
- [13] C. Deffayet, S. Mukohyama and A. Vikman, Phys. Rev. Lett. **128**, 041301 (2022), arXiv:2108.06294; C. Deffayet, A. Held, S. Mukohyama and A. Vikman, JCAP **11**, 031 (2023), arXiv:2305.09631; Phys. Rev. D **112**, 065011 (2025), arXiv:2504.11437.
- [14] T. Damour and A. Smilga, Phys. Rev. D **105**, 045018 (2022), arXiv:2110.11175.
- [15] A. V. Smilga, Int. J. Mod. Phys. A **32**, 1730025 (2017), and references therein.
- [16] D. S. Kaparulin, S. L. Lyakhovich and A. A. Sharapov, Eur. Phys. J. C **74**, 3072 (2014).
- [17] The reduction, its consistency (Gauss law and off-diagonal equations vanish on the ansatz), symmetric criticality, and the auxiliary-field form were verified symbolically; see the code repository.
- [18] S. G. Matinyan, G. K. Savvidy and N. G. Ter-Arutyunyan-Savvidy, Sov. Phys. JETP **53**, 421 (1981).
- [19] G. K. Savvidy, Nucl. Phys. B **246**, 302 (1984).

- [20] M. Lüscher, Nucl. Phys. B **219**, 233 (1983).
- [21] P. Dahlqvist and G. Russberg, Phys. Rev. Lett. **65**, 2837 (1990).
- [22] B. Simon, Ann. Phys. **146**, 209 (1983).
- [23] H. Yoshida, Phys. Lett. A **150**, 262 (1990).
- [24] E. Ott and J. C. Sommerer, Phys. Lett. A **188**, 39 (1994).
- [25] P. Ashwin, J. Buescu and I. Stewart, Nonlinearity **9**, 703 (1996).
- [26] L. Arnold, G. Papanicolaou and V. Wihstutz, SIAM J. Appl. Math. **46**, 427 (1986).
- [27] C. Jarzynski, Phys. Rev. E **48**, 4340 (1993).
- [28] E. Ott, Phys. Rev. Lett. **42**, 1628 (1979).
- [29] A. I. Neishtadt, J. Appl. Math. Mech. **48**, 133 (1984).
- [30] C. Grebogi, S. W. McDonald, E. Ott and J. A. Yorke, Phys. Lett. A **99**, 415 (1983).
- [31] S. Bleher, C. Grebogi and E. Ott, Physica D **46**, 87 (1990).
- [32] B. Müller and A. Trayanov, Phys. Rev. Lett. **68**, 3387 (1992).
- [33] T. Kunihiro, B. Müller, A. Ohnishi, A. Schäfer, T. T. Takahashi and A. Yamamoto, Phys. Rev. D **82**, 114015 (2010).